Hints: Integrals – 3. Differentiation under the integral sign and Frullani integrals

Homework

The hints to each problem are cumulative. Read one, try again, and open the next only if you are still stuck.

Problem 1

Hint 1. Let $F(\alpha) = \int_0^\infty e^{-\alpha x}\frac{\sin\beta x}{x}dx$. Then $F'(\alpha) = -\int_0^\infty e^{-\alpha x}\sin\beta x\,dx = -\frac{\beta}{\alpha^2+\beta^2}$ (uniform convergence for $\alpha \geqslant \alpha_0 > 0$), and $F(\alpha) \to 0$ as $\alpha \to \infty$.

Hint 2. Hence $F(\alpha) = \arctan\frac{\beta}{\alpha}$. Letting $\alpha \to 0^+$ (justified as in the class sheet, Problem 4) gives $\int_0^\infty \frac{\sin\beta x}{x}dx = \frac{\pi}{2}\operatorname{sgn}\beta$.

Problem 2

Hint 1. $F'(\alpha) = \int_0^1 \frac{dx}{(1+\alpha^2x^2)\sqrt{1-x^2}}$, and $F(0) = 0$. Substitute $x = \sin\theta$.

Hint 2. $\int_0^{\pi/2}\frac{d\theta}{1+\alpha^2\sin^2\theta} = \frac{\pi}{2\sqrt{1+\alpha^2}}$ (use $t = \tan\theta$). Integrating, $F(\alpha) = \frac{\pi}{2}\ln\left(\alpha+\sqrt{1+\alpha^2}\right) = \frac{\pi}{2}\operatorname{arsinh}\alpha$.

Problem 3

Hint 1. For $\alpha > 0$: $F'(\alpha) = \int_0^\infty \frac{2\alpha x^2}{(1+\alpha^2x^2)(1+x^2)}dx$. Decompose into partial fractions in x^2: $\frac{x^2}{(1+\alpha^2x^2)(1+x^2)} = \frac{1}{\alpha^2-1}\left(\frac{1}{1+x^2} - \frac{1}{1+\alpha^2x^2}\right)$.

Hint 2. This gives $F'(\alpha) = \frac{\pi}{1+\alpha}$, and $F(0) = 0$, so $F(\alpha) = \pi\ln(1+\alpha)$ for $\alpha \geqslant 0$ (and $\pi\ln(1+|\alpha|)$ in general). Check at $\alpha = 1$: $x = \tan\theta$ gives $-2\int_0^{\pi/2}\ln\cos\theta\,d\theta = \pi\ln 2$.

Problem 4

Hint 1. Look for $f(x,t)$ on $[0,1]$ whose $\partial_t f$ blows up near the corner $(x,t) = (0,0)$. Try $f(x,t) = \frac{xt^3}{(x^2+t^2)^2}$, $f(x,0) = 0$.

Hint 2. Directly, $F(t) = \int_0^1 f\,dx = \frac{t}{2(1+t^2)}$, so $F'(0) = \frac{1}{2}$. But for each fixed x, $\partial_t f(x,0) = \lim_{t\to 0}\frac{xt^2}{(x^2+t^2)^2} = 0$, so the formal answer is 0. The hypothesis that fails: $\partial_t f$ is not continuous on $[0,1]\times[-\delta,\delta]$ (at $x = t$ it equals $\frac{1}{4t}$, unbounded). Another standard example: $\int_0^\infty \frac{\sin\alpha x}{x}dx$, whose formal derivative $\int_0^\infty \cos\alpha x\,dx$ diverges.

Problem 5

Hint 1. This is Frullani with $f(u) = e^{-u}$, $a = 1$, $b = \cosh t$: the answer is $\ln\cosh t$.

Hint 2. The two pieces $\int_0^\infty \frac{e^{-x}}{x}$ and $\int_0^\infty \frac{e^{-x\cosh t}}{x}$ diverge separately, so no splitting is legitimate on $(0,\infty)$. On $[\varepsilon,\infty)$ both converge and their difference is $\int_\varepsilon^{\varepsilon\cosh t}\frac{e^{-u}}{u}du$, which tends to $\ln\cosh t$; this limit is just the improper integral of the whole integrand, so it does not depend on how you split.

Problem 6

Hint 1. Write the integrand as an integral: $\frac{x-1}{\ln x} = \int_0^1 x^y\,dy$ and $\frac{e^{-ax}-e^{-bx}}{x} = \int_a^b e^{-yx}\,dy$. Swap the order (justify by Fubini/Tonelli: the integrands are nonnegative, or absolutely integrable).

Hint 2. First: $\int_0^1 \frac{dy}{y+1} = \ln 2$. Second: $\int_0^\infty e^{-yx}\cos cx\,dx = \frac{y}{y^2+c^2}$, so the answer is $\int_a^b \frac{y\,dy}{y^2+c^2} = \frac{1}{2}\ln\frac{b^2+c^2}{a^2+c^2}$.

Problem 7

Hint 1. $\partial_a I = \int_0^{\pi/2} \frac{2a\cos^2 x}{a^2\cos^2 x + b^2\sin^2 x}dx$. Divide by $\cos^2 x$ and set $t = \tan x$: $\int_0^\infty \frac{2a\,dt}{(a^2+b^2t^2)(1+t^2)}$.

Hint 2. Partial fractions give $\partial_a I = \frac{\pi}{a+b}$, so $I = \pi\ln(a+b) + C(b)$. At $a = b$ the integral is $\pi\ln b$, hence $C = -\pi\ln 2$ and $I = \pi\ln\frac{a+b}{2}$.

Problem 8

Hint 1. Rewrite each power with base e: $2^{\log x} = e^{\log 2\cdot\log x} = x^{\log 2}$; $x^{1/\log x} = e^{\log x/\log x} = e$; $2015^x = e^{x\log 2015}$; and $e^{e^{2016x}+6048x} = e^{e^{2016x}}\left(e^{2016x}\right)^3$.

Hint 2. Answers: $\frac{2^{1+\log 2}}{1+\log 2}$; $e\cdot e = e^2$ (the integrand is the constant e); $\frac{2015^x}{\log 2015}+C$; with $u = e^{2016x}$, $\frac{1}{2016}\int u^2 e^u\,du = \frac{e^u(u^2-2u+2)}{2016} + C$.

Problem 9

Hint 1. Frullani needs only that $f(0)$ and $f(\infty)$ exist, not that $f(\infty) = 0$. Here $f = \arctan$, $f(0) = 0$, $f(\infty) = \frac{\pi}{2}$ (for $a, b > 0$).

Hint 2. The answer is $\left(0 - \frac{\pi}{2}\right)\ln\frac{b}{a} = \frac{\pi}{2}\ln\frac{a}{b}$. Check: differentiating in a gives $\int_0^\infty \frac{dx}{1+a^2x^2} = \frac{\pi}{2a}$.

Problem 10

Hint 1. By parts with $u = \sin^2 x$, $dv = \frac{dx}{x^2}$: the boundary terms vanish and you get $\int_0^\infty \frac{\sin 2x}{x}dx = \frac{\pi}{2}$.

Hint 2. With $F(\alpha) = \int_0^\infty \frac{\sin^2\alpha x}{x^2}dx$: $F'(\alpha) = \int_0^\infty \frac{\sin 2\alpha x}{x}dx = \frac{\pi}{2}$ for $\alpha > 0$ and $F(0) = 0$, so $F(\alpha) = \frac{\pi\alpha}{2}$ (uniform convergence of the derivative integral holds on $[\alpha_0, \infty)$; use continuity at 0).

Hint 3. Parseval: for $f = \mathbf{1}_{[-1,1]}$, $\hat{f}(\omega) = \int f(x)e^{-i\omega x}dx = \frac{2\sin\omega}{\omega}$, and $\int|\hat{f}|^2 d\omega = 2\pi\int|f|^2 dx = 4\pi$. So $\int_{-\infty}^\infty \frac{\sin^2\omega}{\omega^2}d\omega = \pi$ and the requested integral is $\frac{\pi}{2}$.